

Iliass Sijelmassi

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SUMMARY

MSc Data Science & AI candidate (École Polytechnique & HEC Paris, GPA 3.8/4.0), Stanford visiting researcher applying ML to large-scale time-series prediction, and incoming Quantitative Researcher Intern at **WorldQuant**.

EDUCATION

- École Polytechnique** — MSc Data Science & AI (Year 1) GPA: 3.8/4.0 Sep 2024 – Jun 2025
– **Coursework:** Probability & Statistics, Linear Algebra & Regression, Optimization, Deep Learning, Machine Learning.
- HEC Paris** — MSc Data Science & AI (Year 2) Sep 2025 – Jun 2026
– **Quant-relevant coursework:** Machine Learning for Financial Markets, Time Series, Causal Inference, Reinforcement Learning, Interpretability & Algorithmic Fairness, ML Ops & Cloud Engineering.
- INP-ENSEEIH, Toulouse** — Dipl. Eng. in CS, Telecom & Applied Math Sep 2020 – Sep 2023
– **Coursework:** Probability, Statistics, Optimization, RL, Algorithms, High-Performance Computing, Scientific Computing (**MATLAB**).

EXPERIENCE

- Quantitative Researcher Intern** | *WorldQuant (Millennium affiliate), Paris* Oct 2026 – Mar 2027
– Incoming 6-month internship within a Millennium systematic pod; details confidential.
- Visiting Student Researcher** | *Stanford University, Stanford, CA* Mar 2026 – Sep 2026
– First-author study predicting a rare, severe post-surgery cardiac complication in **7,653** patients, combining structured medical records with high-frequency bedside **waveforms** (blood pressures, ECG) from a **~1.5 TB** archive; parsimonious pre-operative model reached **AUROC 0.74**, calibrated to the **1.4%** event rate.
– Designed a leakage-controlled evaluation across four model families (pre-operative data, early-ICU signals, missingness-aware and **FFT** waveform features), benchmarking logistic regression against gradient boosting, random forest and a convolutional baseline under patient-grouped cross-validation with in-fold feature selection; **externally validated on unseen data** from a second hospital system (**AUROC 0.695** on ~11,800 admissions; **0.731** after recalibration).
- Data Scientist Intern** | *Crédit Agricole CIB (CACIB), Paris* Mar 2025 – Sep 2025
– Modeled whether internal audit recommendations would be implemented on time across **200k+** historical records, reaching **F1 $\approx$ 0.75** and improving **ROC-AUC** over the heuristic baseline.
– Designed a dual-model approach — **regularized Logistic Regression** for the coefficient-level interpretability auditors required, plus **XGBoost** for performance; drove feature selection with domain experts and validated robustness against **overfitting on out-of-time data**.
- Software Engineering Intern** | *Infosys – Renault, Paris* Mar 2023 – Oct 2023
– Built backend modules for supply-chain platforms in **Java** and **PostgreSQL**.

QUANTITATIVE RESEARCH & ENGINEERING PROJECTS

- Crypto Perpetuals Alpha Strategy** | *ML for Financial Markets, HEC Paris* 2026
– Ranked **#1 of 13 teams** (**~50 students**) on out-of-sample Sharpe for a dollar-neutral long-short strategy across **411 cryptocurrency perpetual futures** (gross of transaction costs).
– Engineered **379** candidate features — **316** retained after **Spearman IC** filtering and correlation pruning — into a **Ridge + XGBoost** ensemble under a rolling 12-month **walk-forward**, with PCA, scaler, and models refit per window and the IC filter fit once on train and frozen as an explicit anti-leakage choice.
- Leukemia Survival Prediction** — **QRT / ENS Data Challenge** | *Cox PH, Python* 2025
– Predicted overall survival on clinical + genomic data from 24 hospitals (**4,500+ patients**) using a **Cox Proportional Hazards** model; finished **top 12%**.
- Cognitive Alpha** — **Spatial Decision Analytics, FIFA World Cup 2022** | *Personal Project, Python* 2025–2026
– Quantified, for each open-play pass, the gap between the player's decision and the spatially optimal alternative: trained an **Expected Threat (xT)** model via **Markov-chain value iteration** on **126k possession actions**, with a NumPy-vectorised pitch-control engine and a **Streamlit** dashboard over **~5.4 GB** of tracking data.

TECHNICAL SKILLS & ACTIVITIES

Programming: Python (primary), SQL (PostgreSQL), R, **MATLAB**, Bash; C++ / Java (working knowledge)
Quantitative & ML: Probability & Statistics, Time-Series, Survival Analysis, Signal/Alpha Research
Tools: PyTorch, XGBoost, Scikit-learn, NumPy, Pandas, MLflow, Docker, Git, Kafka, FastAPI, Dataiku, Linux
Self-study & Certs: Stochastic calculus & options theory (Hull, Natenberg, Wilmott); Akuna Capital Options 101 & 202; **Jane Street** puzzles
Languages: French (Native), English (Fluent); Dutch, Spanish, Portuguese & Arabic (B2)
Activities: Built and sold a **200k-follower** X/Twitter account to **Polymarket**; private mathematics tutor (3+ years); **Chess** (1700 Elo)